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Polyhedral Graphical Statics
For Funicular Structural Form Finding

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8 - Broader Applications of Polyhedral Graphic Statics

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Summary

In this chapter, we showed the broader application of Polyhedral Graphic statistics in other fields of science and briefly introduced research directions and topics that go beyond the polyhedral limitations of this method. Particularly, we show a research project in which graphical methods were used to analyze the structural pattern of a dragonfly wing. The result was then combined with machine learning methods to generate the structure of a wing of an airplane with enhanced out-of-plane performance. We also visited applications in the design of strut-and-tie structures for referenced concrete and further application in designing multi-material structural components where the direction of the deposition of material is adjusted with respect to the internal force flow to maximize mechanical performance. The application of Polyhedral Graphic Statics was shown in the design of cellular solids and briefly discussed how particular subdivision of the force diagram can control the stress distribution in the system and the overall behavior of the structure from bending dominant to stretching dominant system. We also showed the application of the

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structures designed using Polyhedral Graphic Statics in self-healing structural components and 3D-printed structural systems with maximized surface area and minimized mass. Another important topic was the extension of the methods of Polyhedral Graphic Statics to non-polyhedral systems using disjointed force polyhedra. In the end, advanced topics related to completeness, being, and kinematics in Polyhedral Graphic Statics were discussed, which opened the door to many further research directions in this field.

Keywords

analyzing natural structures | dragonfly wings | strut and Tie Method | cellular to shellular geometry | self-healing metal structures | multi-layered sheet-based structures | 3d printed structures | non-polyhedral structures | advanced topics | completeness | bending | kinematics

Information

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